

Quantum 2 HW3

Daniel Haim Breger, 316136944

Technion

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1 Simple Model for Quantization of the Electromagnetic Field

1. Use Maxwell's equations to write time development equations for $\mathcal{E}$ and $\mathcal{B}$.
2. Express $U(t)$ using $V, \mathcal{E}(t), \mathcal{B}(t), \mu_0, \varepsilon_0$. Use

$$\int_V d\mathbf{r} \sin^2 kz = \int_V d\mathbf{r} \cos^2 kz = \frac{V}{2}$$

3. We'll define $\omega = ck$ and

$$\chi(t) = \sqrt{\frac{\varepsilon_0 V}{2\hbar\omega}} \mathcal{E}(t) \qquad \Pi(t) = \sqrt{\frac{V}{2\mu_0\hbar\omega}} \mathcal{B}(t)$$

show that the equations of motion for $\chi(t)$ and $\Pi(t)$ are identical in their form to those of X and P in subsection 1.

1.1 Time Development

The relevant Maxwell's equations for time development are

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t} \qquad \nabla \times \mathbf{B} = \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) \quad (1)$$

and since no current or charges at all other than out waves were specified, we assume the current $\mathbf{J} = 0$. Plugging the given fields into the equations we find

$$k\mathcal{E} = \frac{\partial \mathcal{B}}{\partial t} \qquad k\mathcal{B} = -\mu_0\varepsilon_0 \frac{\partial \mathcal{E}}{\partial t} \quad (2)$$

we can take the derivative of the equations and plug them into the other ones to find 2nd order differential equations

$$\frac{\partial^2 \mathcal{E}}{\partial t^2} + \frac{k^2}{\mu_0\varepsilon_0} \mathcal{E} = 0 \qquad \frac{\partial^2 \mathcal{B}}{\partial t^2} + \frac{k^2}{\mu_0\varepsilon_0} \mathcal{B} = 0 \quad (3)$$

1.2 Energy in Time

$$U(t) = \int_V d\mathbf{r} \left(\frac{\varepsilon_0}{2} \mathbf{E}^2 + \frac{1}{2\mu_0} \mathbf{B}^2 \right) \quad (4)$$

$$= \int_V d\mathbf{r} \left(\frac{\varepsilon_0}{2} \mathcal{E}^2 \sin^2(kz) + \frac{1}{2\mu_0} \mathcal{B}^2 \cos^2(kz) \right) \quad (5)$$

$$= \frac{V}{2} \left(\frac{\varepsilon_0}{2} \mathcal{E}^2 + \frac{1}{2\mu_0} \mathcal{B}^2 \right) \quad (6)$$

1.3 Same Form

The equations of motion for $\mathcal{E}$ and $\mathcal{B}$ are 2nd order linear ODEs, which are invariant to multiplication by scalars. The transformation of parameters from $\mathcal{E}$ and $\mathcal{B}$ to χ and Π simply multiplies by a scalar, therefore the equations of motion will remain the same.

2 Quantization of the Field

1. Express $\hat{H}_F$ in terms of $\hat{a}$ and $\hat{a}^\dagger$ and the number operator $\hat{N} = \hat{a}^\dagger \hat{a}$.
2. Calculate the expectation values $\langle \hat{E}(\mathbf{r}) \rangle_n$, $\langle \hat{B}(\mathbf{r}) \rangle_n$, $\langle \hat{H}_F \rangle$ in state $|n\rangle$.

2.1 Field Hamiltonian

From the definitions for the photon creation and annihilation operators we can find expressions for χ and Π using the operators:

$$\hat{\chi} = \frac{1}{\sqrt{2}} (\hat{a} + \hat{a}^\dagger) \qquad \hat{\Pi} = \frac{1}{\sqrt{2}i} (\hat{a} - \hat{a}^\dagger) \quad (7)$$

therefore

$$\hat{H}_F = \frac{\hbar\omega}{2} (\hat{\chi}^2 + \hat{\Pi}^2) \quad (8)$$

$$= \frac{\hbar\omega}{4} ((\hat{a} + \hat{a}^\dagger)^2 - (\hat{a} - \hat{a}^\dagger)^2) \quad (9)$$

$$= \frac{\hbar\omega}{4} (\hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} + \hat{a}^{\dagger 2} - \hat{a}^2 + \hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a} - \hat{a}^{\dagger 2}) \quad (10)$$

$$= \frac{\hbar\omega}{2} (\hat{a}\hat{a}^\dagger + \hat{N}) \quad (11)$$

$$= \frac{\hbar\omega}{2} (\hat{a}^\dagger\hat{a} - \hat{a}^\dagger\hat{a} + \hat{a}\hat{a}^\dagger + \hat{N}) \quad (12)$$

$$= \frac{\hbar\omega}{2} (2\hat{N} + [\hat{a}, \hat{a}^\dagger]) \quad (13)$$

$$= \frac{\hbar\omega}{2} (2\hat{N} + 1) \quad (14)$$

$$= \hbar\omega \left(\hat{N} + \frac{1}{2} \right) \quad (15)$$

2.2 Expectation Values

First we notice

$$\langle \hat{E}(\mathbf{r}) \rangle_n = \langle n | \hat{E} | n \rangle \quad (16)$$

which if we look at the definition of the creation and annihilation operators which define $\hat{E}$, the result of the above equation is always zero, therefore

$$\langle \hat{E}(\mathbf{r}) \rangle_n = 0 \quad (17)$$

and the same holds for $\langle \hat{B}(\mathbf{r}) \rangle_n$. For $\hat{H}_F$:

$$\langle n | \hat{H}_F | n \rangle = \hbar\omega \langle n | \left(\hat{N} + \frac{1}{2} \right) | n \rangle = \hbar\omega \left(n + \frac{1}{2} \right) \quad (18)$$

3 Coherent States of the Electromagnetic Field

1. Show that the state $|\alpha\rangle$ is a normalized eigenstate of the annihilation operator a . Find the corresponding eigenvalue. Calculate the expectation value $\langle N \rangle_\alpha$ in this state.
2. Given $|\psi(t=0)\rangle = |\alpha\rangle$, show that

$$|\psi(t)\rangle = \exp\left(-i\frac{\omega t}{2}\right) |\alpha e^{-i\omega t}\rangle$$

3. Calculate the expectation values $\langle \mathbf{E}(\mathbf{r}) \rangle_t$ and $\langle \mathbf{B}(\mathbf{r}) \rangle_t$ in the state $|\psi(t)\rangle$. Assume α is real.
4. Show that $\langle \mathbf{E}(\mathbf{r}) \rangle_t$ and $\langle \mathbf{B}(\mathbf{r}) \rangle_t$ satisfy Maxwell's equations.
5. Calculate the classical energy of the field for $\mathbf{E}_{cl} = \langle \mathbf{E}(\mathbf{r}) \rangle_t$ and $\mathbf{B}_{cl} = \langle \mathbf{B}(\mathbf{r}) \rangle_t$ and compare with $\langle H_F \rangle_t$.
6. Assume $|\alpha| \gg 1$ and justify the statement that the states $|\alpha\rangle$ have a "classical" character.

4 Question 4

1. What is the hamiltonian to 0-th order, and how will the states $|g\rangle, |e\rangle$ develop?
2. What will the states of $|g\rangle, |e\rangle$ be after a half-pi pulse?
3. For an initial state $|g\rangle$, after a half-pi pulse, we'll turn the electric field off and let the system develop in time. What is the state now?
4. After the free propagation we'll send another half-pi pulse. What will the state be after this? What is the likelihood to find the atom in the excited state ($P_e(\Delta)$)?
5. We can measure $P_e(\Delta)$ and tune ω using it. The sensitivity is defined by $S \equiv \left| \frac{dP_e}{d\Delta} \right|$.
 - (a) What is the meaning of this value?
 - (b) What is our clock's sensitivity?
 - (c) When will we get maximum sensitivity?
 - (d) What is our error range?
6. Estimate how long until the clock will be 1 second off.

4.1 0-th Order Hamiltonian

Given that $\Omega \gg \Delta$, The 0th order hamiltonian is just the element that doesn't include Δ

$$H = \frac{1}{2} \hbar \Omega \sigma_x \quad (19)$$

and to find the time development of the states, since the hamiltonian is time independent

$$U(t) = e^{-\frac{i}{\hbar} H t} = e^{-\frac{i}{2} \Omega t \sigma_x} \quad (20)$$

$$= \cos\left(\frac{\Omega t}{2}\right) \mathbb{1} - i \sin\left(\frac{\Omega t}{2}\right) \sigma_x \quad (21)$$

$$= \cos\left(\frac{\Omega t}{2}\right) (|g\rangle \langle g| + |e\rangle \langle e|) - i \sin\left(\frac{\Omega t}{2}\right) (|g\rangle \langle e| + |e\rangle \langle g|) \quad (22)$$

Therefore

$$|g(t)\rangle = U(t) |g\rangle \quad (23)$$

$$= \cos\left(\frac{\Omega t}{2}\right) |g\rangle - i \sin\left(\frac{\Omega t}{2}\right) |e\rangle \quad (24)$$

$$|e(t)\rangle = U(t) |e\rangle \quad (25)$$

$$= \cos\left(\frac{\Omega t}{2}\right) |e\rangle - i \sin\left(\frac{\Omega t}{2}\right) |g\rangle \quad (26)$$

4.2 After Half-Pi Pulse

The states will be

$$|g'\rangle = \cos\left(\frac{\pi}{4}\right) |g\rangle - i \sin\left(\frac{\pi}{4}\right) |e\rangle \quad (27)$$

$$= \frac{1}{\sqrt{2}} |g\rangle - \frac{i}{\sqrt{2}} |e\rangle \quad (28)$$

$$|e'\rangle = \cos\left(\frac{\pi}{4}\right) |e\rangle - i \sin\left(\frac{\pi}{4}\right) |g\rangle \quad (29)$$

$$= \frac{1}{\sqrt{2}} |e\rangle - \frac{i}{\sqrt{2}} |g\rangle \quad (30)$$

4.3 Free Development

When the electric field is turned off, the hamiltonian is

$$H = \frac{1}{2} \hbar \Delta \sigma_z \quad (31)$$

so the time development operator is

$$U_2(t) = e^{-\frac{i}{\hbar} H t} = e^{-\frac{1}{2} i \Delta \sigma_z} \quad (32)$$

$$= \cos\left(\frac{\Delta t}{2}\right) \mathbb{1} + i \sin\left(\frac{\Delta t}{2}\right) \sigma_z \quad (33)$$

therefore

$$|g''\rangle = U_2(t) |g'\rangle \quad (34)$$

$$= \cos\left(\frac{\Delta T}{2}\right) |g'\rangle + i \sin\left(\frac{\Delta T}{2}\right) (|e\rangle \langle e| - |g\rangle \langle g|) |e'\rangle \quad (35)$$

$$= \cos\left(\frac{\Delta T}{2}\right) |g'\rangle + i \sin\left(\frac{\Delta T}{2}\right) (|e\rangle \langle e| - |g\rangle \langle g|) \left(\frac{1}{\sqrt{2}} |e\rangle - \frac{i}{\sqrt{2}} |g\rangle\right) \quad (36)$$

$$= \cos\left(\frac{\Delta T}{2}\right) |g'\rangle + i \sin\left(\frac{\Delta T}{2}\right) \left(\frac{1}{\sqrt{2}} |e\rangle + \frac{i}{\sqrt{2}} |g\rangle\right) \quad (37)$$

$$= \frac{1}{\sqrt{2}} \left(\cos\left(\frac{\Delta T}{2}\right) (|g\rangle - |e\rangle) + i \sin\left(\frac{\Delta T}{2}\right) (|e\rangle + |g\rangle) \right) \quad (38)$$

$$= \frac{1}{\sqrt{2}} \left(\left(\cos\left(\frac{\Delta T}{2}\right) + i \sin\left(\frac{\Delta T}{2}\right) \right) |g\rangle - \left(\cos\left(\frac{\Delta T}{2}\right) - i \sin\left(\frac{\Delta T}{2}\right) \right) |e\rangle \right) \quad (39)$$

$$= \frac{1}{\sqrt{2}} \left(e^{\frac{i\Delta t}{2}} |g\rangle - i e^{-\frac{i\Delta t}{2}} |e\rangle \right) \quad (40)$$

4.4 Another Pulse

We are back to using the electric field time propagator, therefore

$$|g'''\rangle = U\left(\frac{\pi}{2\Omega}\right) |g''\rangle \quad (41)$$

$$= \left(\cos\left(\frac{\pi}{4}\right) (|g\rangle \langle g| + |e\rangle \langle e|) - i \sin\left(\frac{\pi}{4}\right) (|g\rangle \langle e| + |e\rangle \langle g|) \right) \frac{1}{\sqrt{2}} \left(e^{\frac{i\Delta t}{2}} |g\rangle - ie^{-\frac{i\Delta t}{2}} |e\rangle \right) \quad (42)$$

$$= \frac{1}{\sqrt{2}} \left(\cos\left(\frac{\pi}{4}\right) \left(e^{\frac{i\Delta t}{2}} |g\rangle - ie^{-\frac{i\Delta t}{2}} |e\rangle \right) - i \sin\left(\frac{\pi}{4}\right) \left(e^{\frac{i\Delta t}{2}} |e\rangle - ie^{-\frac{i\Delta t}{2}} |g\rangle \right) \right) \quad (43)$$

$$= \frac{1}{2} \left(\left(e^{\frac{i\Delta t}{2}} - e^{-\frac{i\Delta t}{2}} \right) |g\rangle - i \left(e^{\frac{i\Delta t}{2}} + e^{-\frac{i\Delta t}{2}} \right) |e\rangle \right) \quad (44)$$

$$= \sinh\left(\frac{i\Delta t}{2}\right) |g\rangle - i \cosh\left(\frac{i\Delta t}{2}\right) |e\rangle \quad (45)$$

$$= i \sin\left(\frac{\Delta t}{2}\right) |g\rangle - i \cos\left(\frac{\Delta t}{2}\right) |e\rangle \quad (46)$$

therefore

$$P_e(\Delta) = |\langle e | g''' \rangle|^2 \quad (47)$$

$$= \cos^2\left(\frac{\Delta t}{2}\right) \quad (48)$$

4.5 Tuning and Sensitivity

4.5.1 Meaning

The meaning of the sensitivity S is how strongly a deviation from the resonant frequency affects the operation of the clock. A high value means even small deviations will cause significant changes to the probability of the states flipping.

4.5.2 Our Clock's Sensitivity

$$S = \left| \frac{dP_e(\Delta)}{d\Delta} \right| = \left| t \sin\left(\frac{\Delta t}{2}\right) \cos\left(\frac{\Delta t}{2}\right) \right| = \left| \frac{t}{2} \sin(\Delta t) \right| \quad (49)$$

4.5.3 Maximum Sensitivity

We can see the maximums will be found at $t = \frac{(2k+1)\pi}{2\Delta}$

4.5.4 Error Range

$$\delta\Delta = \frac{3\pi - \pi}{2T} = \frac{\pi}{T} \quad (50)$$

4.6 Time Until 1s Error

Using the given equation

$$\delta t = \frac{\delta f}{f_0} t \implies t = \frac{f_0 \delta t}{\delta f} \quad (51)$$

therefore

$$t = \frac{9\,192\,631\,770 \text{ Hz} \cdot 1 \text{ s}}{\pi \text{ s}^{-1}} = 92.72 \text{ yr} \quad (52)$$